Regression Project
Due Friday May 22

Description of Dataset
The Royal Draconic Conservatory of the kingdom of Draconia, studies dragons and related species living in a protected reserve.
One characteristic that the Conservatory measures is the heat of a Dragon's flame, which is widely used in applications such as metal working.

Each dragon undergoes a standardised flame test, and the resulting measurement is recorded as its **Flame Heat Output (FHO)**, measured in degrees Celsius.

While FHO depends on a dragon's physical and magical characteristics, it is also influenced by environmental variation and measurement noise. Your goal is to build a regression model to predict FHO based on observable features of dragons.

You are given the following variables:

- FHO: The target variable — Flame Heat Output (in degrees Celsius).
- WSP: Wingspan (in meters).
- MASS: Mass (in metric tons).
- AGE: Age (in years).
- SPD: Sustained flight speed (in km/h).
- HID: Hide thickness (in cm).
- SPC: Species (e.g. Dragon, Wyvern or Hyrdra).

1. **Single Variable Regression:**
   Fit single-variable regression models to predict FHO from each of the following:
   (a) WSP
   (b) MASS
   (c) AGE
   (d) SPC

   For each model:
   - Report the training R^2.
   - Evaluate performance on a test set.
   - Produce appropriate plots (e.g. scatter plots with fitted lines).

   Comment on which variables appear to be most informative. [15]

2. **Multiple Variable Regression:**
   Build multiple regression models using combinations of the available variables.
   - Experiment with different subsets of variables.
   - Justify your modelling choices.
   - Report both training and test performance.

Which models perform best, and why? [15]

3. **Model Comparison:**
   Compare the models you have constructed.

   - Does adding variables improve performance?
   - Are there signs of overfitting?
   - Which model would you recommend, and why?

   [10]

4. **Interpretability:**
   Interpret the coefficients of your preferred model.

   - Do the signs and magnitudes of coefficients make sense?
   - Which features have the strongest influence on FHO?
   - Are there any surprising results?

   Explain your answers in the context of dragon biology and behaviour. [15]

5. **Model Evaluation:**
   Use appropriate tools to evaluate your models.

   - Residual plots
   - Error analysis
   - AIC/BIC or cross-validation (if familiar)

   Discuss the reliability and generalisation of your models. [15]

6. **Feature Engineering:**
   Construct new features to improve your model.

   - Consider transformations (e.g. logarithms, square roots).
   - Consider interaction terms between variables.

   Explain your choices and whether they improved performance.

   Hint: Some characteristics may only be informative when considered together. [20]